PHOTONICS



# High-Speed Open Silicon Photonics Reference Designs


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# Architecture can close on paper. Manufacturing readiness cannot.

- Manufacturing of Photonic-Electronic systems requires supply chain collaboration
- Open Silicon reference designs build better understanding between players
- Flex and Lightmatter joined forces to build a test vehicle for manufacturability



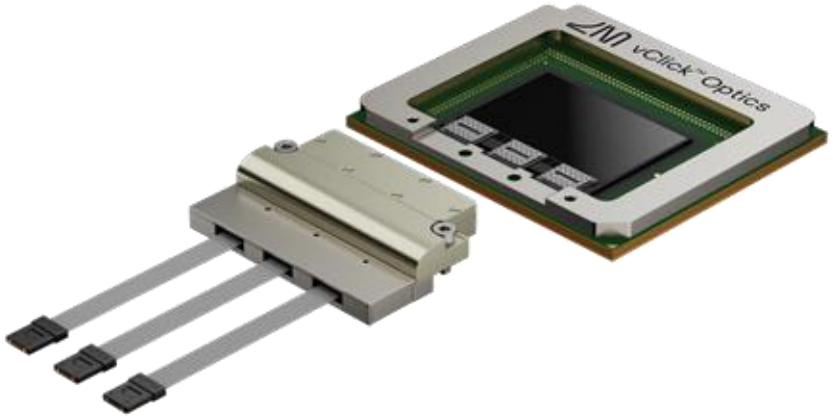
# Tolerances Span Six Orders of Magnitude

## Foundry



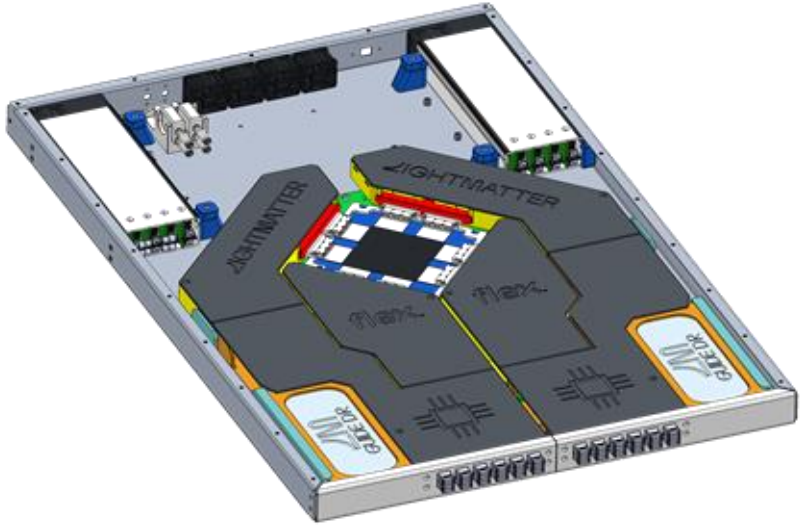
Sub-micron

## OSAT




Microns

## Assembly




Tens of microns to meters

Intertwined domains: electrical, optical, thermal & mechanical DFX; virtual/physical validation

A hair’s width held across a football field.

# Where CPO Assembly Gets Hard

**CPO Package** → **CPO Assembly** → **OE Connection** → **Fiber Routing** → **System Integration**

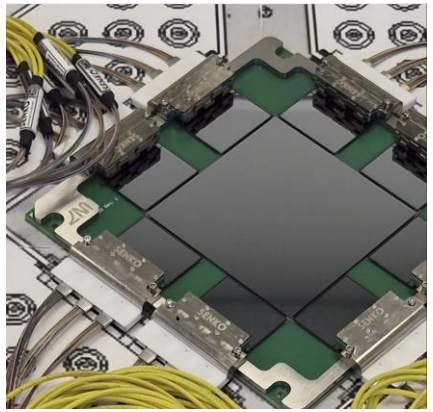
- Logic, memory, XPU
- OE

- Precision BGA attach
- TSA, Warpage.

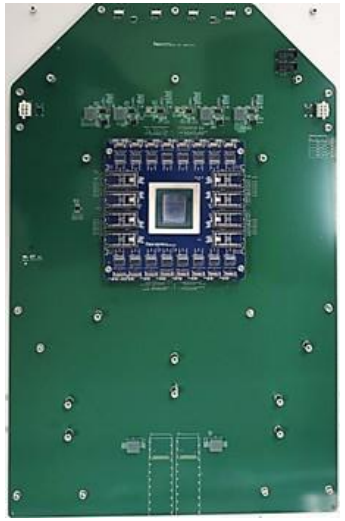
- FAU/socket Alignment
- FAU Insertion forces

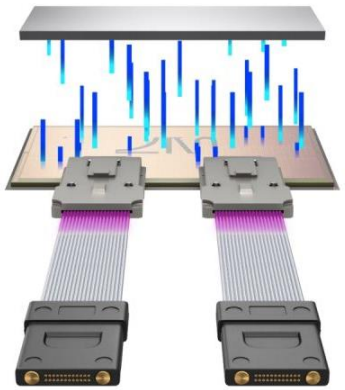
- Optical link integrity (fiber bend, alignment)
- Clearances

- Optical signal integrity
- Connectorization
- Cooling

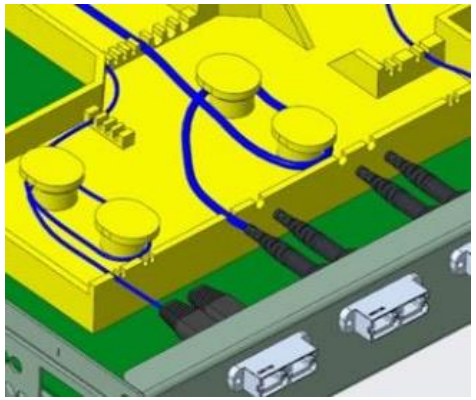


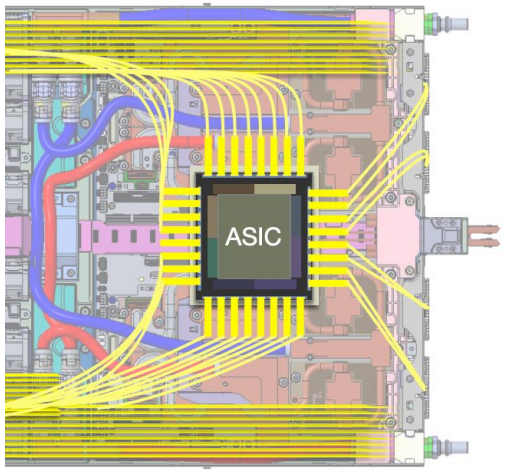
From Foundry/OSAT



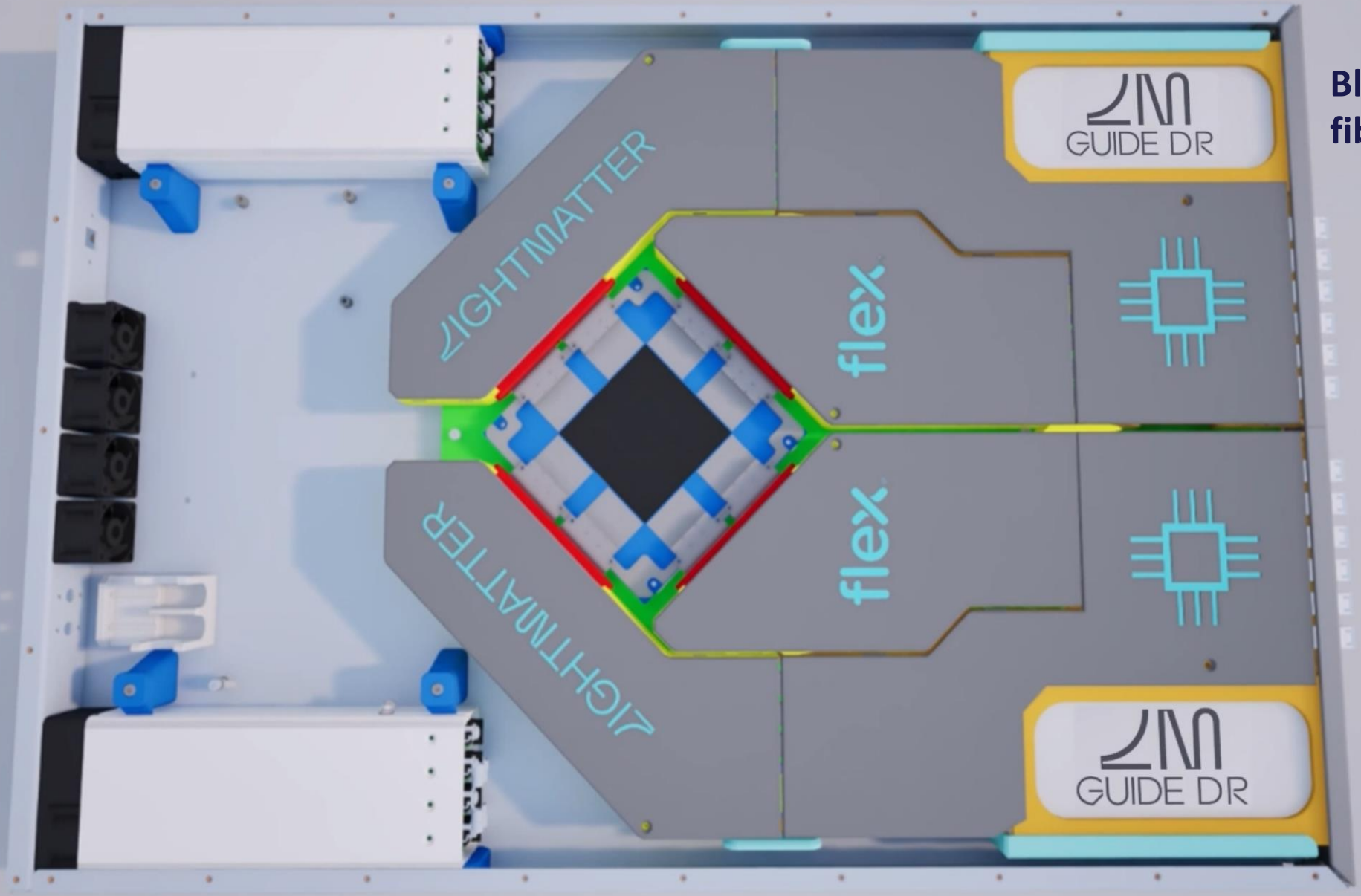


FAU connections must occur after reflow





# Manufacturability, Serviceability, Automation




Gang connected FAUs

Blind-connect fiber trays

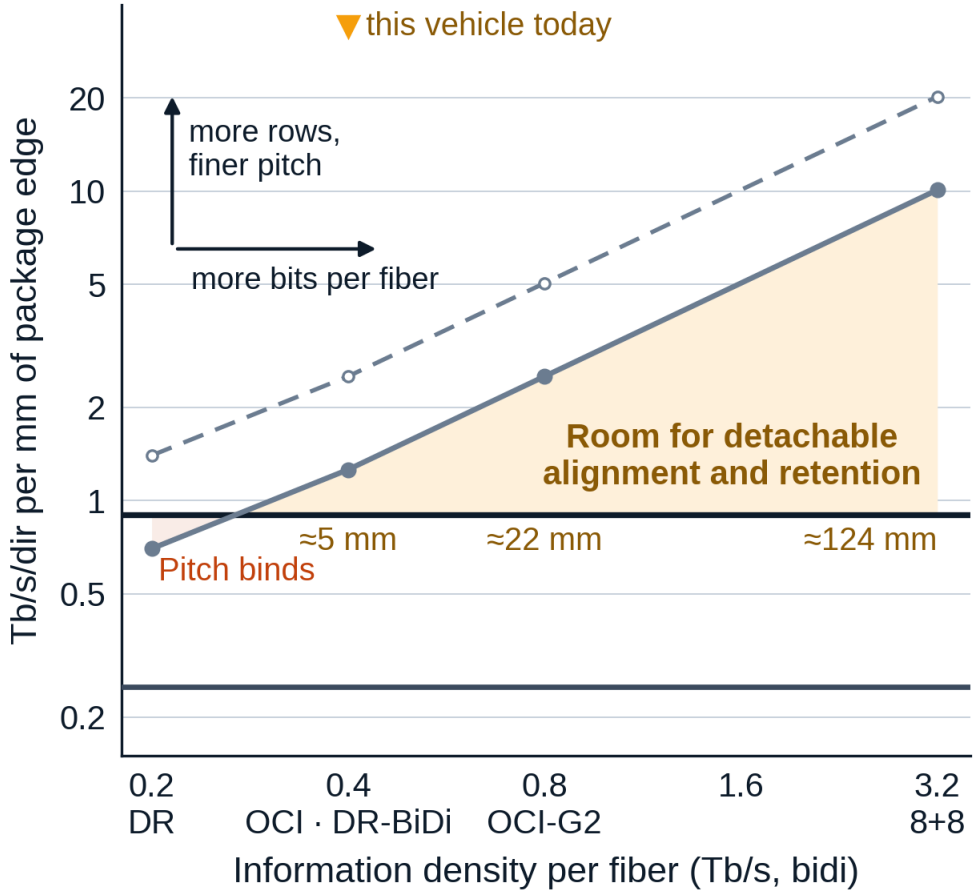
Configurable front panel

Blind-connect modular lasers

ORv3 system tray

# Fiber Density Pays for Detachability

## escape per mm of package edge, NPO vs CPO

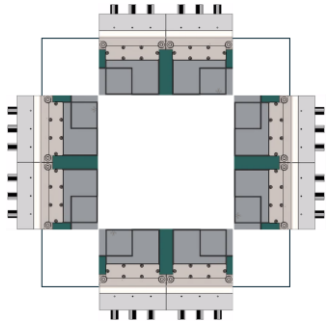



Fiber array, vertical, 2 rows

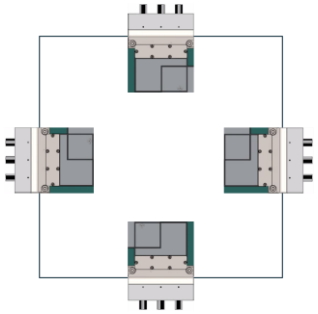
Fiber array, edge, 1 row, 127 µm

**CPO silicon edge (switch-side)**

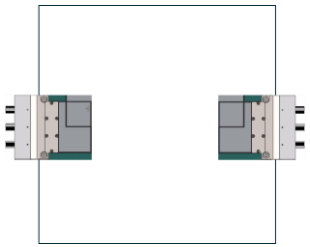
NPO, socketed NPO: copper escape



**8 dFAUs**
Most bandwidth
768 data fibers
154–614 Tb/s/dir



**4 dFAUs**
Easier to manufacture
384 data fibers
77–307 Tb/s/dir

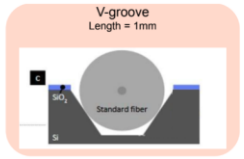


**2 dFAUs**
Simple to service
192 data fibers
38–154 Tb/s/dir

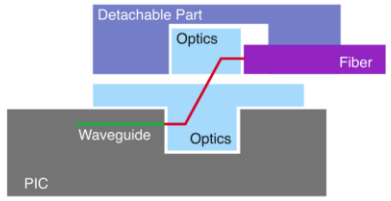
Switch-side edge densities per OIF EEI: CPO 2.0, copper 1.0 Tb/s/mm bidi, 0.9 and 0.5 usable. Fiber counts include laser feeds. Labels under the dark line: edge left for alignment and retention structures per 96-fiber dFAU, one row; corners ignored.

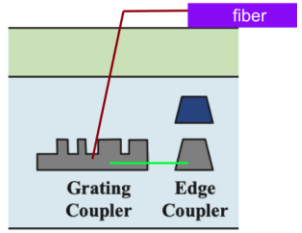
# Passive Alignment: HVM Criteria

Key Performance Indicators for HVM




V-Groove with fixed FAU



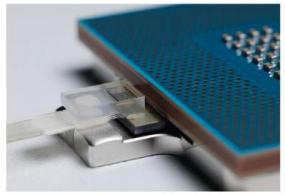

Wafer Level Optical coupler (GC or cavity based)

| Key performance indicator | Edge · 1 row · fixed | Vertical · rows · detachable |
|---|---|---|
| High yield | Partial | Yes |
| Meets product performance | Yes | Yes |
| High throughput (UPH) | Partial | Yes |
| Known-good OE | No | Yes |
| Scalable across package architectures | No | Yes |
| Serviceability | No | Yes |
| **Package edge per Tb/s (slide 7)** | Yes: one row, no hardware | Partial: paid by fiber density |

HVM readiness needs partnership across equipment vendors, materials suppliers, OSATs and manufacturers.

# Passive Alignment: Measured Results

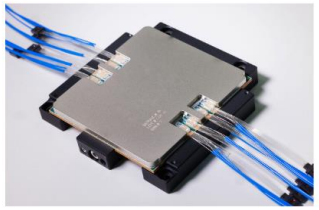
## Product/Platform Kits (PPK)





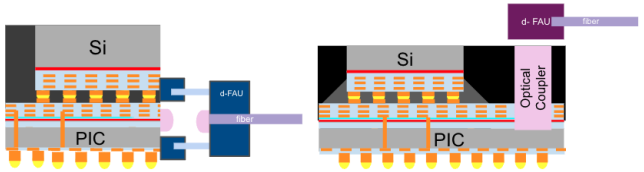
- 2D MCP w/ V-grooves
- 56G NRZ, 112G PAM4, 224G PAM4
- 1 dB mean facet Insertion Loss
- 181 FAU with 99.5% yield at <2 dB
- Package Test = 99.5% (500+ units built to date)

## Multi-Reticle Assembly test Vehicle (ATV)



- 4000 mm^2 3D chip on wafer multi-reticle with Edge Attach
- 1.5 dB mean facet Insertion Loss
- 62 FAU built to date with 95% Optical attach yield at <2 dB

## Single Reticle ATV



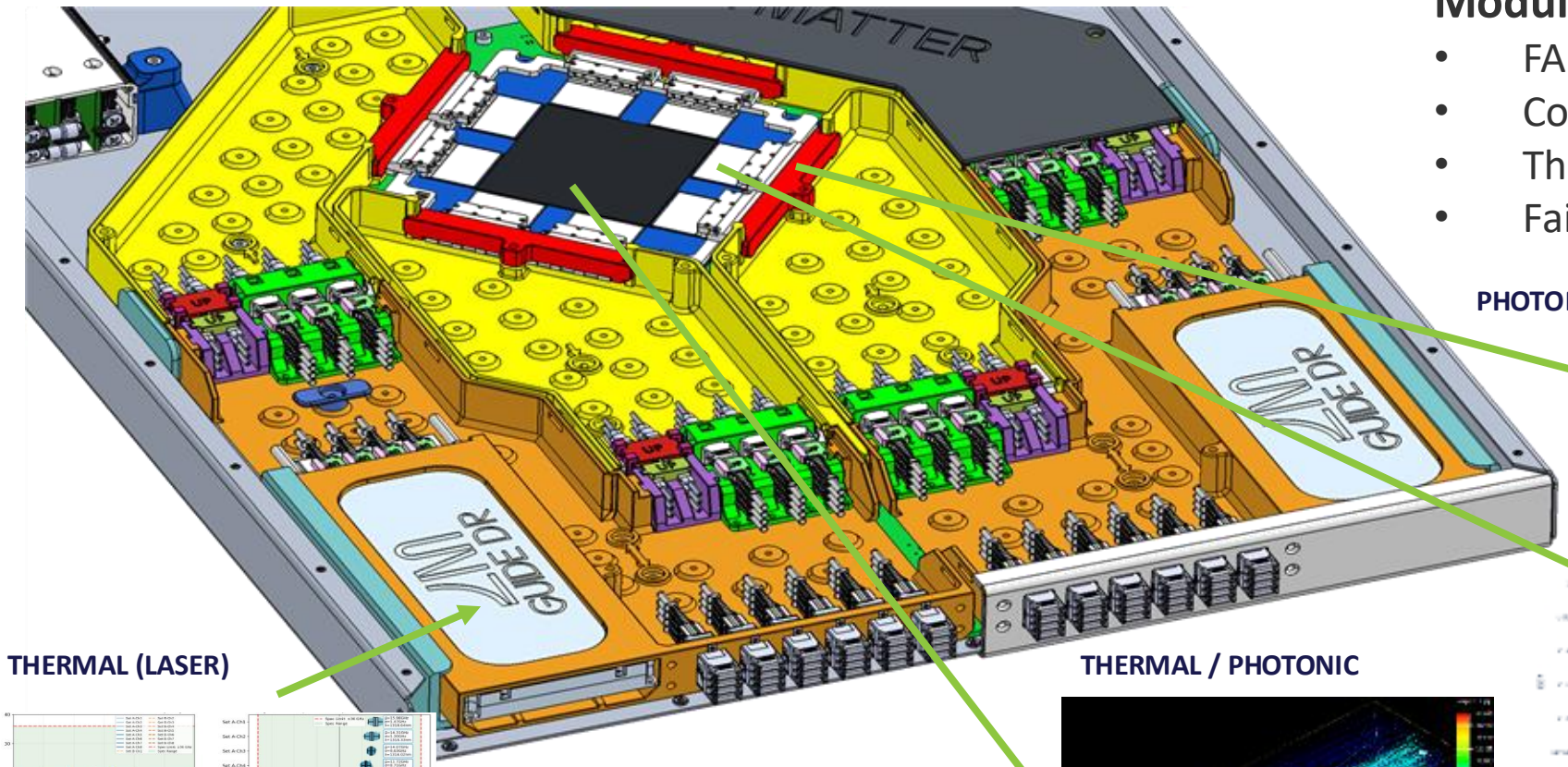

- Full reticle 3D chip stack
- Demonstrate IL<1.5dB at high yields (>99%) (based on simulations)

ATV: Assembly test vehicle

*Silicon Technology Definition*

Results from detachable Fiber Attach Units very positive and yields similar to socketed high rate electrical connectors.

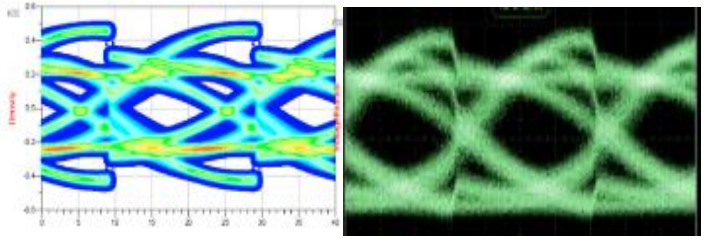
# Testing CPO with Detachable FAUs



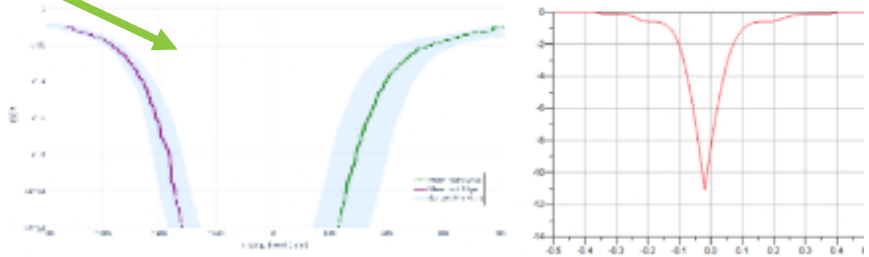
**Modular, sub-system level tests**

- FAU insertion effectiveness
- Configurable optical path
- Thermal effects on link
- Failure-chain tracing

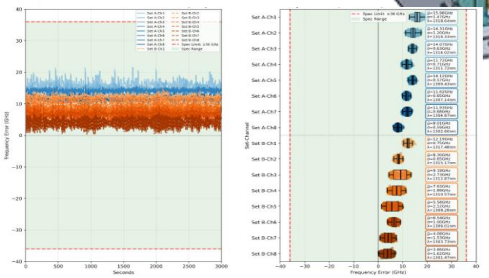
**PHOTONIC**



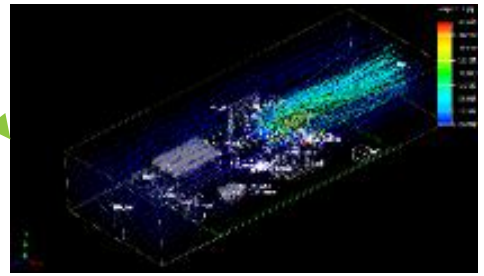
**ELECTRICAL**



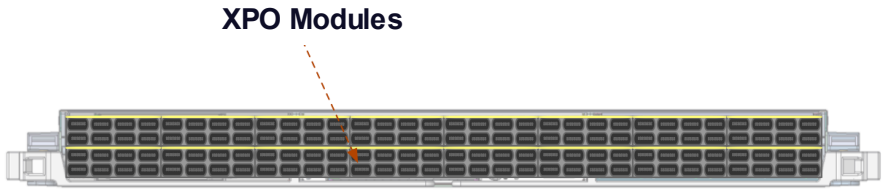
**THERMAL (LASER)**

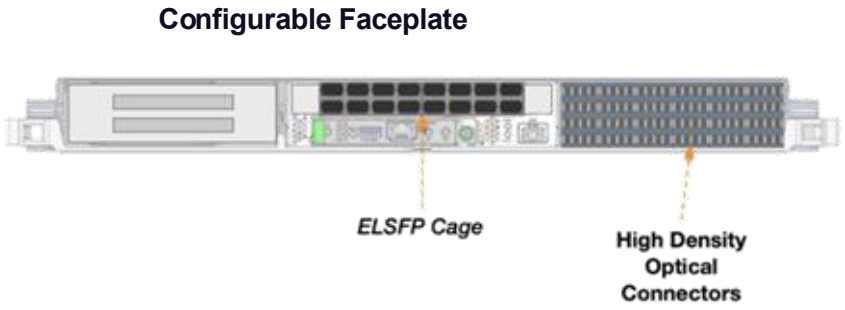


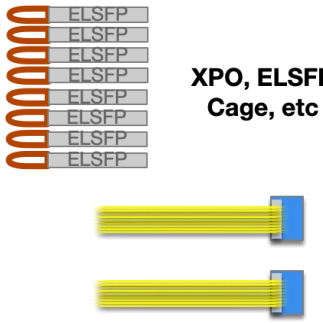
**THERMAL / PHOTONIC**

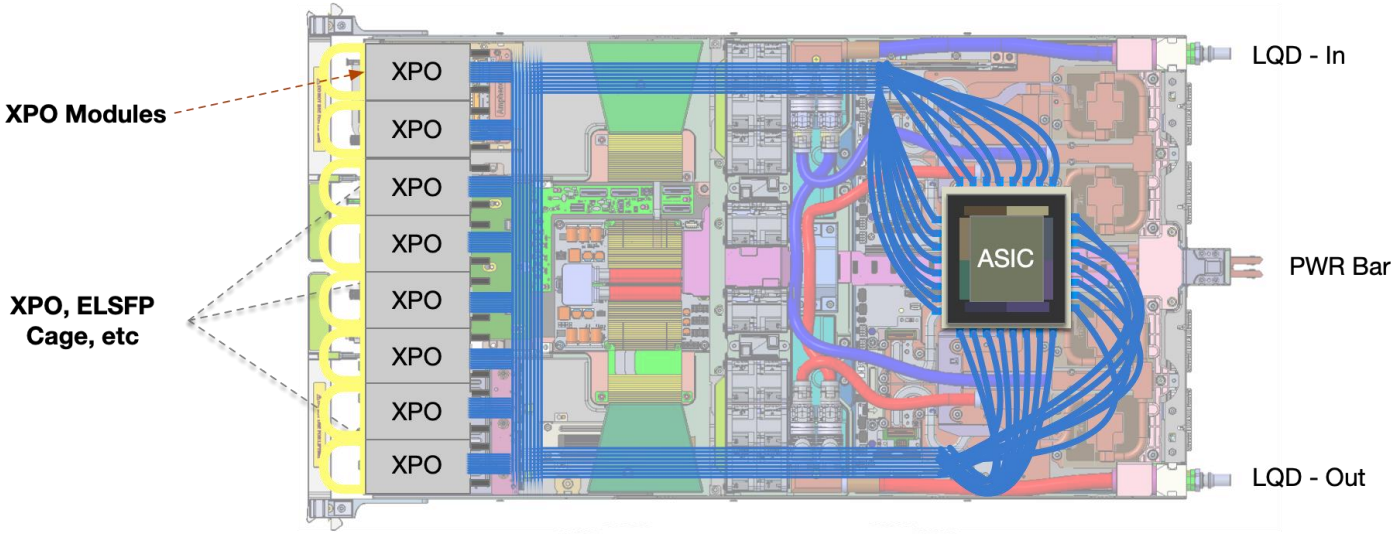


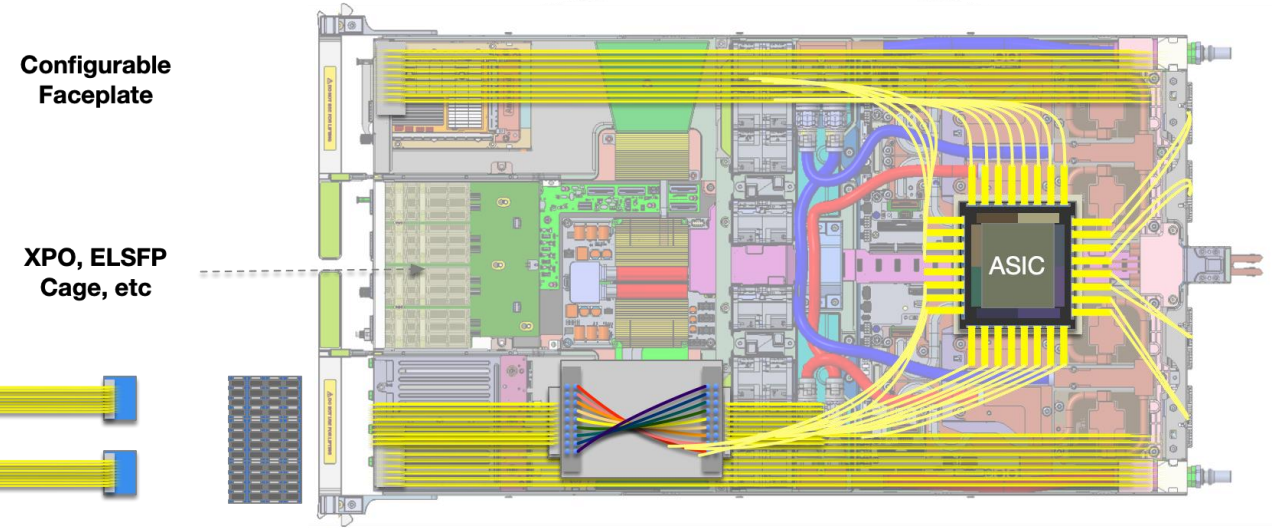
# Change One Variable at a Time



# What We Learn at Each Boundary




| CPO Package | CPO Assembly | OE Connection | Fiber Routing | System Integration |
|---|---|---|---|---|
| **Link margin vs. budget** | **Warpage · BGA attach yield** | **FAU yield · UPH · alignment window · IL · mating repeatability** | **Bend radius · clearance** | **Laser blind-mate · tray thermals · signal integrity** |
| ***→ Optical reference design*** | ***→ Mechanical + thermal reference design*** | ***→ Optical + mechanical reference design*** | ***→ Mechanical reference design*** | ***→ Electrical + thermal reference design*** |

**Each finding becomes a row of the tray contract.**

# Standardize the Contract, Compete on Implementation

| Dimension | FPO (front-panel) | NPO (near-package) | CPO (co-packaged) |
|---|---|---|---|
| **Signaling: optical** | Pluggable optics at the faceplate; fiber at the cage | OE on tray; fiber egress to faceplate connector | Detachable FAU; passive alignment; blind-mate fiber; ELS laser path |
| **Signaling: electrical** | Host SerDes to cage (LR-class) | Short-reach host trace (XSR/VSR); unretimed to half-retimed | Unretimed XSR-class; no connector in the channel |
| **Mechanical** | Cage + faceplate envelope; pluggable service | OE keep-outs; fiber egress zones; socketed / mezzanine FRU variants | Package envelope; blind-mate fiber; ELS shelf; FRU = the package |
| **Power** | Cage power limits | OE rails on tray; sequencing | ELS + OE package power; sequencing |
| **Thermal** | Faceplate airflow | Shared airflow, OE + ASIC | Package-level cooling; liquid-ready |
| **Management** | CMIS at the cage | CMIS + tray sensor extension | RAS telemetry via DC-SCM path |

# Call to Action

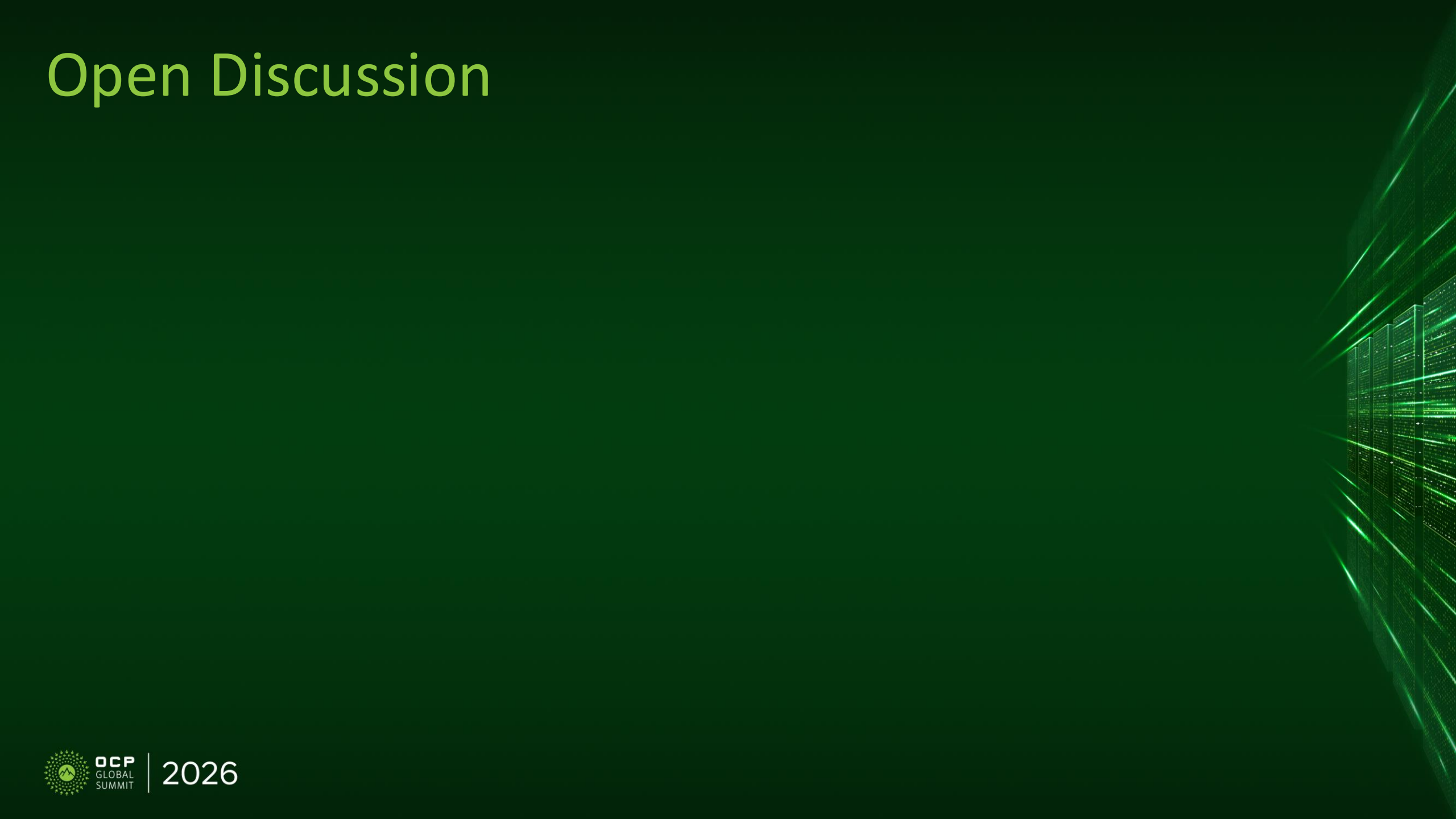
**Read**
and comment on the Open Silicon Photonics for AI Systems whitepaper. (See QR)

**Join**
OCP MHS Interconnect-for-AI workstream — biweekly calls.
bijan@lightmatter.co

**Help define**
Link architecture · laser/RAS · mechanical integration · thermal/packaging.

- Problem to solve: the missing layer is a contract, not a component — an open platform contract for photonics-ready MHS trays.
- Timeline: whitepaper available now; specification drafts through 2027.
- Yesterday: the tray contract behind these designs (Tue Oct 13, Room 212).
- The spec is ours to co-author.

# Open Discussion

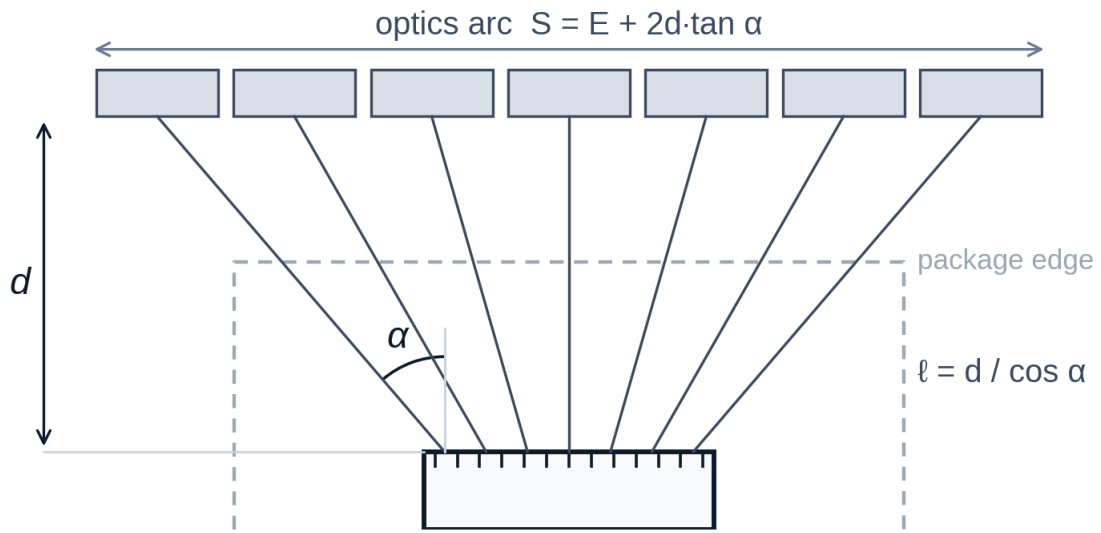


# Backup

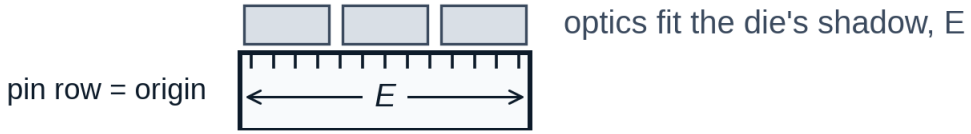


# Distance Buys Optics Room and Costs Reach

**NPO d = tens of mm · package edge escapes copper, 0.25 Tb/s/dir per mm**




**CPO d ≈ 0 · die edge escapes light, 0.9 Tb/s/dir per mm**




**Optics arc per edge:** $S = E + 2d \cdot \tan \alpha$

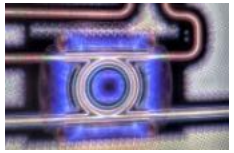
**Electrical channel:** $\ell = d / \cos \alpha$

**Package-edge escape:** CPO 0.9 · NPO 0.25 Tb/s/dir per mm

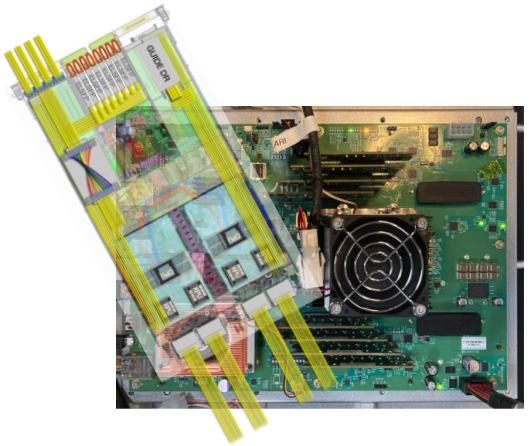
| | NPO: d = tens of mm | CPO: d ≈ 0 |
|---|---|---|
| **Escape** | Copper throat; the wall stays | Light at the die edge; wall moves out ~3.6× |
| **Optics room** | Arc grows with d; detachable hardware is cheap | Die's shadow; needs fiber density |
| **Channel** | ℓ grows with d: loss, layers, may retime | mm-scale, unretimed, no connector |
| **Thermal, service** | Decoupled from the ASIC; each OE is a FRU | Beside the ASIC; FRU = the package |

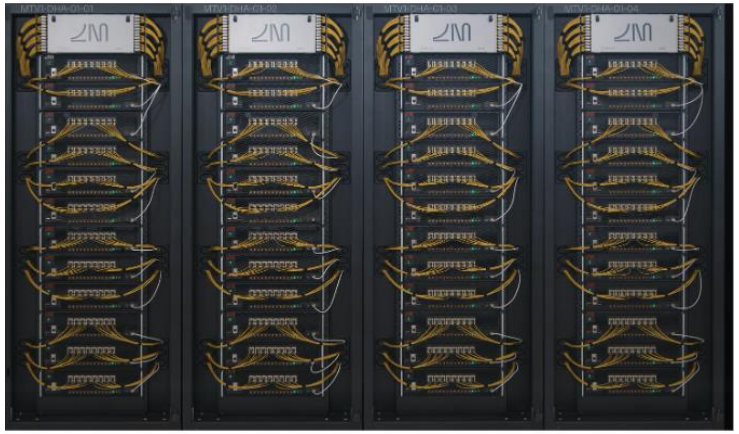
Plan view, one routing layer, square die: corners and extra layers change the constants, not the law. Switch-side edge densities per OIF EEI. E = pin-row length; d = distance from the pin row; α = fan-out angle.

# Reliability at Every Scale









**Device / Wafer**

- Thermal Stress
- Electrical Stress
- Optical Stress
- Reliability Modeling
- Lifetime Analysis

**Package**

- Temp/Humidity
- Mechanical Stress
- Optical Stress

**System**

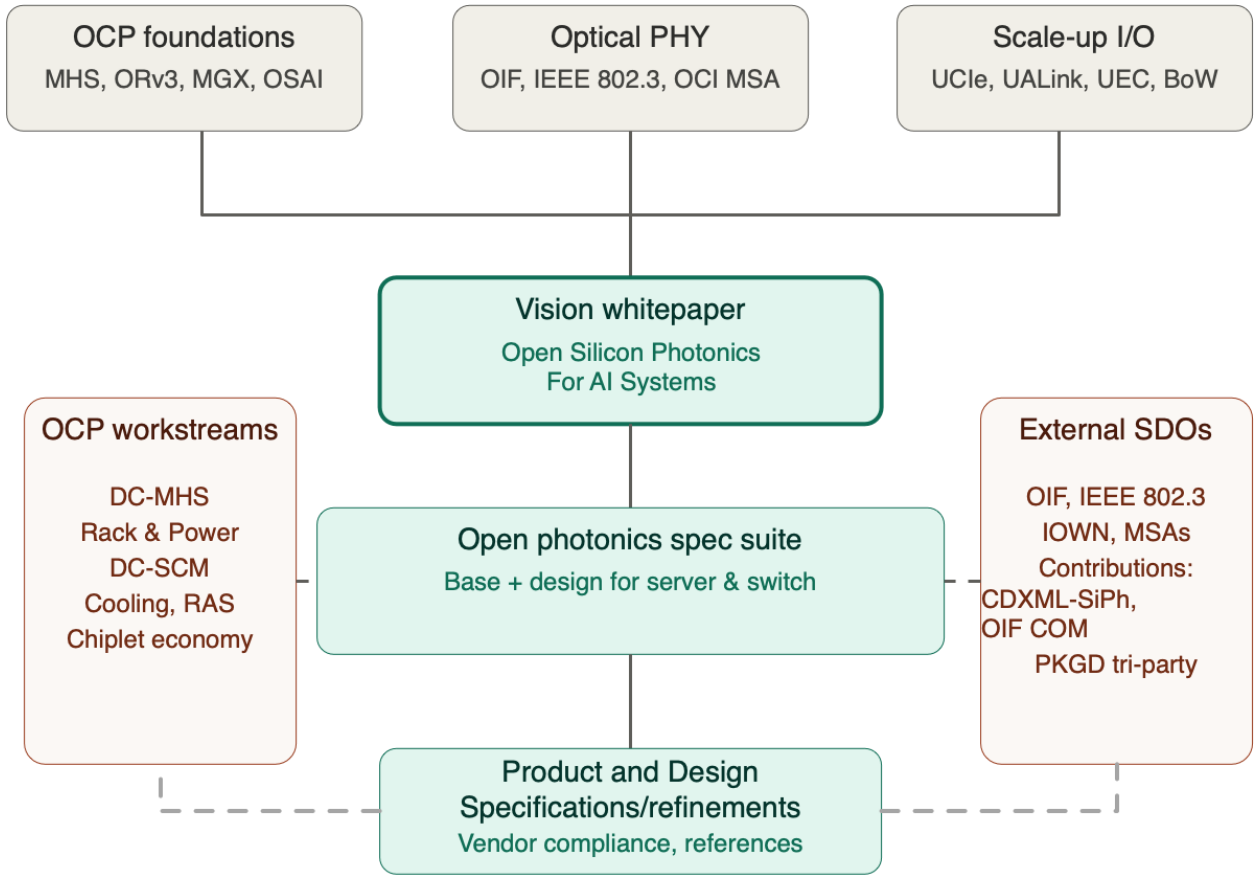
- HTOL test
- Performance

**Racks**

- Extended Operation
- Thermal management
- FIT Lifecycle

# Where Reference Designs Land

**Mechanical, power, signaling, thermal and management contracts.**




Solid arrow = contribution / dependency  Dashed arrow = coevolution / feedback

# An Open Community Vision

LIGHTMATTER

flex

Celestica™

hyve solutions®

CORNING

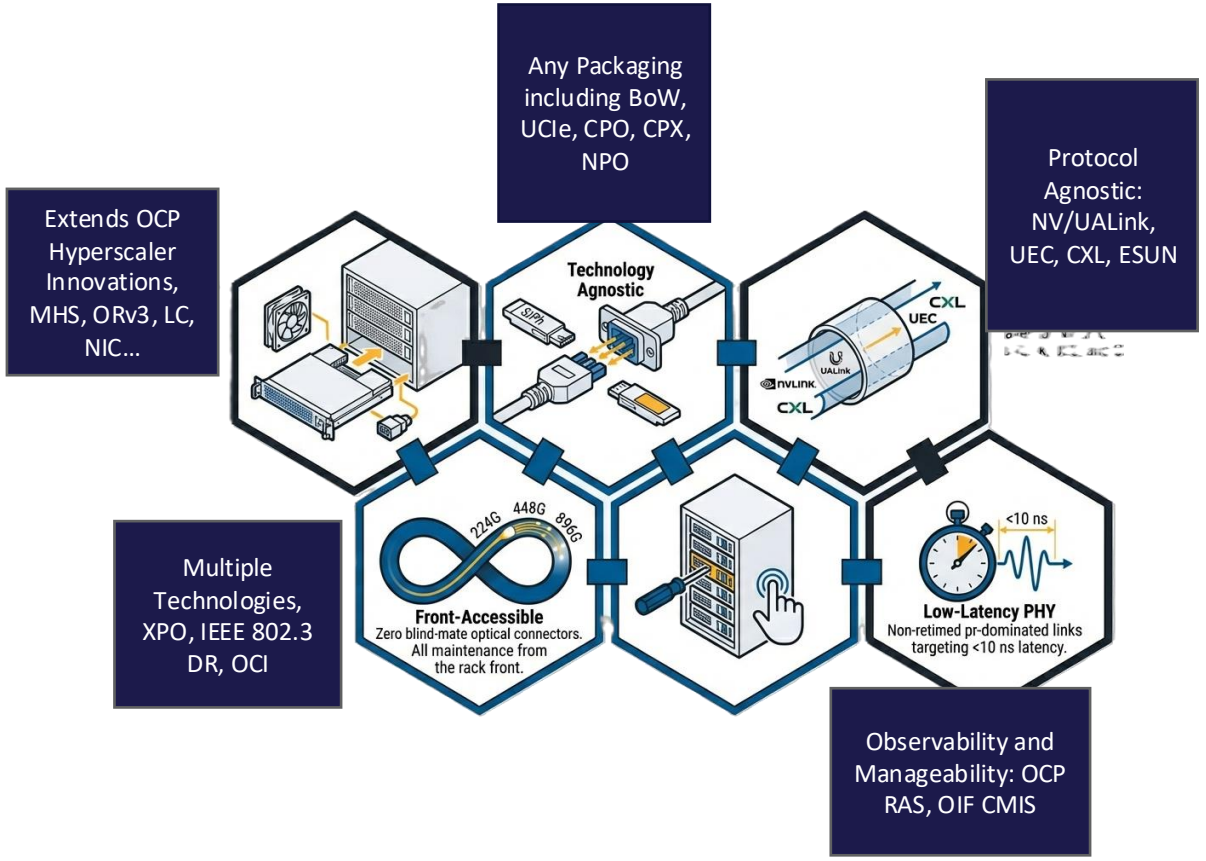
KEYSIGHT

FOXCONN

QCT™

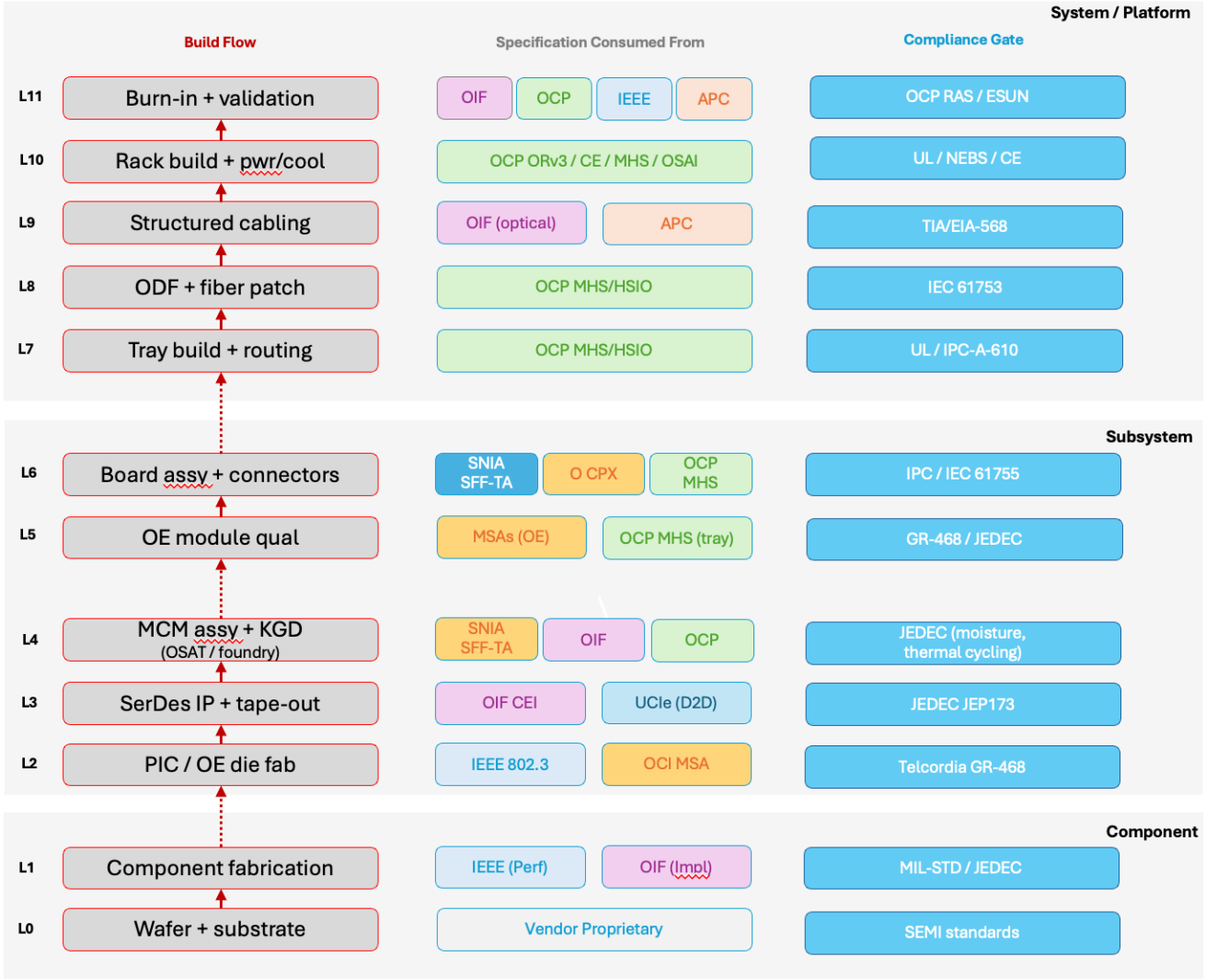
DELL Technologies

Qualcomm




Announced March 16, 2026

# Requirements and Steps




**Compliance gates are orthogonal**: they don't define function, they gate shipment

**Schedule risk concentrates at L4** (JEDEC qual) **and L10** (UL/NEBS certification)

# Towards Open Reference Hardware

2025
**Framework**
Escape-geometry lens and six-axis taxonomy developed

H1 2026
**Rev 0.1 and RC series**
Release candidates through RC8 — review across OIF, IEEE, UCIe, adjacent MSAs

Q3 2026
**Server review**
Presented to the OCP Server project; comment period

Aug 10
**Whitepaper Release 1.x *published***
Steering Committee review period completes and the contributed Architecture Vision published


OPEN Compute Project

ARCHITECTURE VISION: OPEN SILICON PHOTONICS FOR AI SYSTEMS

A Vision for Scaling AI Compute and Switching to Next-Generation Density Through Open Photonic Interconnect

*1.0.3*

Authors:

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Mohammad Alavirad, Dell Technologies
Jian Chen, Dell Technologies
Jiayin Ma, Flex

Date: JULY, 2026

Aug 12
**OCP APAC**
The implementation era starts where the systems are built

Q4 2026
**Global Summit**
Base Specification momentum the critical next deliverable

